

SECTION 13065-A FAN FILTER UNITS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

1.2 RELATED WORK

- A. This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for fan filter units:
 - 1. Division 1 sections included in the project specifications.
 - 2. Drawings and the contract
- B. CAUTION: Use of this Section without including all of the above-listed items will result in omission of basic requirements
- C. In the event of conflict regarding the requirements for painting of the referenced equipment, vessels, and piping between this Section and any other section, the provisions of this Section shall govern.

1.3 QUALITY ASSURANCE

- A. Units shall be products of manufacturer regularly engaged in production of such units and issuing complete catalog data on such products.
- B. Manufacturer shall assure that all equipment provided does not exceed maximum allowable vibration peak-to-peak displacement data listed in the latest ASHRAE Systems and Applications Handbook.
- C. Manufacturer shall assure that all equipment provided does not exceed maximum allowable sound power levels that are consistent with the project criteria of NC 60 at a minimum external available pressure of 0.4 inch W.G. (100 Pa). Maximum Allowable Sound Power Levels (PWL) for a 2' x 4' Fan Filter Unit (dB re 10^{-12} Watt) in Octave Bands (Hz):

Criterion Octave Band Center Frequency (Hz)

Space Type	63	125	250	500	1K	2K	4K
Ballroom NC-60	74	67	62	57	55	53	51
Bay/Chas	78	71	65	60	58	55	53

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e NC-60							

- D. Measure sound power using sound intensity techniques integrated, by scanning across the ULPA or HEPA filter of the unit. The sound intensity methodology is defined in ISO/DIS 9614-2 (1994 Draft International Standard). The test methodology must be approved by the engineer prior to the test.
- E. Vibration
 - 1. With the unit supplying 100 feet per minute HEPA face velocity, the average of vertical acceleration measured at four corners with system supported on soft spring (minimum 3-inch static deflection) should not exceed 75 dB re 1 μ g (gravity) in any octave band between 8 and 63 Hz.

1.4 SUBMITTALS

- A. Submit shop drawings.
- B. Submit with shop drawings, operating point plotted on curves and vibration peak-to-peak displacement data.
- C. Submit manufacturer's printed installation instructions.
- D. Submit operating and maintenance manuals.
- E. Submit installation and start-up certification.
- F. Submit outlet octave band sound power data at operating condition.

1.5 JOB CONDITIONS

- A. Do not operate fans for any purpose, temporary or permanent, until housing is clean, filters are in place, non-sealed bearings have been lubricated, and fan has been run under observation.

PART 2 - PRODUCTS

2.1 FANS

- A. Equivalent fan selections shall not increase motor horsepower, increase noise level, increase tip speed by more than 10 percent, or increase inlet air velocity by more than 20 percent, from that scheduled.

- B. Base fan performance on altitude rating conditions.
- C. Refer to Drawings for performance requirements.

2.2 FILTERS

- A. Each fan filter unit shall include ULPA filters rated at 99.9995% efficient at 0.12 microns.

2.3 MOTORS AND CONTROLS

- A. Motors: Provided with equipment. Motors shall be provided with VFD speed control with the ability to accept remote signal for speed control.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. By the Cleanroom contractor.

3.2 STARTUP

- A. By the Cleanroom contractor.

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